Date

Practise Questions,

3.4,

3.4(1a) $x + 2y = 4$
$3x - 5y = 1$

So,

$$Ax = B$$

$$\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

And,

$$A = (1)(-5) - (2)(3)$$
$$= -11$$

Also,

For, A^{-1}.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \frac{1}{-11}\begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}$$

Then,

$$X = A^{-1}B$$
$$= \frac{1}{-11}\begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

So,

$$(-5)(4) + (-2)(1) = -22$$

And,

$$(-3)(4) + (1)(1) = -11$$

At last,

$$X = \frac{1}{-11}\begin{bmatrix} -22 \\ -11 \end{bmatrix}$$

$$\therefore x = 2, y = 1,$$

Date

1.b) $5x + y = 13$
$3x + 2y = 5$

Sol^n:

$$\begin{bmatrix} 5 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 13 \\ 5 \end{bmatrix}$$

Then,

$$\Delta = 5(2) - 1(3) = 7$$

Also,

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix}$$

So,

$$X = A^{-1}B$$
$$= \frac{1}{4} \begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 13 \\ 5 \end{bmatrix}$$

$R_1 = 26 - 5 = 21$
$R_2 = -39 + 25 = -14$

$$\therefore X = \frac{1}{7} \begin{bmatrix} 21 \\ -14 \end{bmatrix}$$

$\therefore x = 3, y = -2$

Date

2.a.) $x+y+z=6$
$2x+3y+z=10$
$x-y+2z=7$

Here,

Subⁿ eqⁿ ① from ②:

$2x+3y-z-x-y-z$
$x+2y=4$

Subⁿ eqⁿ ① from ③:

$(x-y+2z)-(x+y+z)$
$-2y+z=1$

Then,

$x+2y=4$
$-2y+z=1$

From, 2nd,

$x=1+2y$

Subᵗ,

$x+y+(1+2y)=6$
$x+3y=5$

So,

$x=5-3y$

So,

$5-3y+2y=4$
$y=1$

Then,

$x=2$
$z=3$

Date

b) $x + 2y + z = 9$
$2x + y + 3z = 13$
$3x + 2y + 2z = 16$

Sol^n,

$2x + 2y + 3z - (2x + 4y + 2z)$
$-3y + z = -5$

Also,

$3x + 2y + 2z - (3x + 6y + 3z)$
$-4y - z = -11$

And,

$-3y + z = -5$
$-4y - z = -11$

Add,

$-7y = -16$
$\therefore y = \frac{16}{7}$

Then, $-3(16/7) + z = -5$
$\therefore z = \frac{13}{7}$

Also,

$x + 2(16/7) + 13/7 = 9$

$\therefore x = \frac{18}{7}$

Date

1.c) $3x + 2y = -2$
$x + 4y = 6$

Sol^n,
Mul^n eq^n ② by 3
$3x + 12y = 18$

Then,
Sub^t eq^n ①
$(3x + 12y) - (3x + 2y) = 18 - (-2)$
$10y = 20$
$\therefore y = 2$

Also,
$x + 4(2) = 6$
$\therefore x = -2$

1.d) $2x + 3y = 6$
$x - y = 1$

Sol^n, Mul eq^n ② by 2
$2x - 2y = 2$

Then, sub^t ①,
$(2x + 3y) - (2x - 2y) = 6 - 2$
or, $5y = 4$

$\therefore y = \frac{4}{5}$

so, $x = \frac{9}{5}$

Date

So,

$$2(z-2)+(z-1)+z=5$$
$$4x-5=5$$
$$z=\frac{5}{2}$$

Then, $y=3/2$ and $x=\frac{1}{2}$

e.) $5x+3y+9z=-1$
$-2x+3y-z=-2$
$-x-4y+5z=1$

Here,

mul ③ by 2

$$-2x-8y+10z=2$$

Now,

$$(-2x-8y+10z)-(-2x+3y-z)$$
$$-11y+11z=4$$

Div by 11,

$$\therefore y-z=-\frac{4}{11}$$
$$\therefore y=z-\frac{4}{11}$$

So, subst ②,

$$-2x+3y-z=-2$$
$$-2x+3\left(z-\frac{4}{11}\right)-z=-2$$
$$-2x+2z-\frac{12}{11}=-2$$

mul by 11,

$$-22x+22x-12=-22$$
$$\cancel{x}\ x-z=\frac{5}{11}$$
$$\therefore x=\frac{5}{11}+z$$

Date

Then,

Sub into eqn (1)

$$5x + 3y + 9z = -1$$

$$5\left(z + \frac{9}{11}\right) + 3\left(z - \frac{4}{11}\right) + 9z = -1$$

or, $5z + \frac{25}{11} + 3z - \frac{12}{11} + 9z = -1$

or, $17z + \frac{13}{11} = -1$

Mult by 11,

$$187z + 13 = -11$$

or, $187z = -24$

$$\therefore z = -\frac{24}{187}$$

Then,

$$y = z - \frac{4}{11}$$

or, $y = -\frac{24}{187} - \frac{68}{187}$

$$\therefore y = \frac{-92}{187}$$

So, $x = z + \frac{5}{11}$

$$= -\frac{24}{187} + \frac{85}{187}$$

$$= \frac{61}{187}$$

Date

2c.) $x+y+z=3$
$2x-y+z=3$
$3x+y-z=3$

Soln,

$x-2y=0$
$\therefore x=2y$

Also,

$2x-2z=0$
$\therefore x=z$

Then,

$2y+y+2y=3$
$5y=3$
$\therefore y=\frac{3}{5}$

Then, $x=6/5$

$z=6/5$

d.) $2x+y+z=5$
$x+2y+z=6$
$x+y+2z=7$

Soln,

$x-y=-1$

Also,

$y-z=-1$

And,

$y=z-1$

Then,

$x-(z-1)=-1$
$x=z-2$

Date

b.) $f(x,y,z) = x^3z - 2y^2x + 5z$

Soln,

$$\frac{df}{dx} = x^3z - 2y^2x + 5z$$
$$= 3x^2z - 2y^2$$

Also, $$\frac{df}{dy} = x^3z - 2y^2x + 5z$$
$$= -4xy$$

And, $$\frac{df}{dz} = x^3z - 2y^2x + 5z$$
$$= x^3 + 5$$

So, $\Delta = (3x^2z - 2y^2, -4xy, x^3+5)$

So.

At $(-1, 1, 4)$,

$$3(1)^2(-4) - 2(1)^2 = -12 - 2 = -14$$
$$-4(-1)(1) = 4$$
$$1^3 + 5 = 6$$

$$\Delta f = (-14, -4, 6)$$

At $(0, 1, 0)$

$$3(0)^2(0) - 2(1)^2 = -2$$
$$-4(0)(1) = 0$$
$$0^3 + 5 = 5$$

$$\Delta f = (-2, 0, 5)$$

Date

4.3.3

1.) $f(x,y) = x^2 + y^2$ at:

Soln

$$\frac{df}{dx} = 2x$$

Also $$\frac{df}{dy} = 2y$$

So, $\nabla = (2x, 2y)$

Then,

at $(0,0)$; $\Delta f = (0,0)$

at $(1,3)$; $\Delta f = (2\cdot 1, 2\cdot 3)$
$= (2, 6)$

at $(-1,-5)$ $\Delta f = (-2, -10)$

Date

So,

At, (0,0)

$m = 0$

(4,0)

$m = -16 + 16$
$= 0$

(0,5)

$m = -25 + 30$
$= 5$

(4,5)

$m = -16 - 25 + 16 + 30$
$= 5$

So, At max, 13 at (2,3)
min, 0 at (0,0) and (4,0)

Date

2.) $g(x,y) = xy + x^2 - 4y + 6$ at $x : [-2, 2]$, $y : [1, 3]$

Here,

$$\frac{dg}{dx} = y + 2x$$

$$\frac{dg}{dy} = x - 4$$

$$\Delta g = (y + 2x, x - 4)$$

So, C.P.

$$x - 4 = 0$$

$$\therefore x = 4$$

Since, 4 is outside $[-2, 2]$ so, no c.p.

Now,

At, $(-2, 1)$ $\quad g = (-2)(1) + 4 - 4 + 6$
$= 4$

$(-2, 3)$ $\quad g = (-6) + 4 - 12 + 6$
$= -8$

$(2, 1)$ $\quad g = 2 + 4 - 4 + 6$
$= 8$

$(2, 3)$ $\quad g = 6 + 4 - 12 + 6$
$= 4$

So, At max, 12 at $(-2, 3)$
min, -14 at $(-2, 4)$

Date

At $(-3,-2,1)$

$3(9)(3), -2(4) = 27-8 = 19$

$-4(-3)(-2) = -24$

$(-3)^3+5 = -27+5 = -22$

$\Delta f = (19, -24, -22)$

~~8) $m(x,y) =$~~

4.3.3.2,,

1.) $m(x,y) = x^2 - y^2 + 4x + 6y$ interval $x: [0,4]$ and $y: [0,5]$

Here,

$$\Delta m(x,y) = \left(\frac{\partial m}{\partial x}, \frac{\partial m}{\partial y}\right)$$

$$\frac{\partial m}{dx} = -2x+4$$

$$\frac{\partial m}{dy} = -2y+6$$

so, $\Delta m(x,y) = (-2x+4, -2y+6)$

Also,

Critical Point,

$\Delta = 0$,

$-2x+4=0$

$x=2$

$-2y+6=0$

$y=3$

$\therefore$ CP $= (2,3)$

Also,

$m(2,3) = -4-9+8+18$

$= 13$

Date

4.) $h(x,y) = 2x^2 + 3y^2 - 12x - 18y + 20$ at $x: [1,5]$, $y: [0,4]$

Here,

$$\frac{dh}{dx} = 4x - 12$$

$$\frac{dh}{dy} = 6y - 18$$

So,

C.P, $4x - 12 = 0$

$x = 3$

$6y - 18 = 0$

$\therefore y = 3$

$\therefore$ C.P $= (3,3)$

Then,

$h(3,3) = 18 + 27 - 36 - 54 + 20$

$= -25$

At,

$(1,0)$ $\quad h = 2 - 12 + 20$

$= 10$

$(5,0)$ $\quad h = 50 - 60 + 20$

$= 10$

$(1,4)$ $\quad h = 2 + 48 - 12 - 72 + 20$

$= -14$

$\therefore$ At, max, 10 at $(1,0)$

min, ~~at (3,3)~~ -25 at $(3,3)$

Date

3.) $P(x,y) = x^3 - 3xy + 4$ at $x: [0,2]$, $y: [0,3]$

Here,

$$\frac{dp}{dx} = 3x^2 - 3y$$

$$\frac{dp}{dy} = -3x$$

So, C.P, $-3x = 0$

$\therefore x = 0$

sub,

$3(0)^2 - 3y = 0$

$\therefore y = 0$

So, C.P $= (0,0)$

Then,

At, $(0,0)$ $\quad p = 4$

$(2,0)$ $\quad p = 8 + 4$

$= 12$

$(2,3)$ $\quad p = 8 - 18 + 4$

$= -6$

So,

At max, 12 at $(2,0)$

min, -6 at $(2,3)$